Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Bengaluru, India

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Summary

MLOps Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, building SageMaker ML pipelines and the Terraform and CI/CD that run them. Graduated the internal MLOps SME Program by building a SageMaker pipeline for medical image classification, since published by AWS as an official sample: clinical quality and fairness gates, blue/green canary endpoints with auto-rollback, and PSI drift detection that starts retraining, all in Terraform + CodePipeline. Deep IaC and CI/CD foundation from DevOps engagements at RWS, State Street, and DTCC. Built LeetCode Rating Predictor and fixed a train/serve skew that had tripled its backtest error (MAE 38.76 to 12.46 on 57 contests). Open-source contributor to Feast, Apache Airflow, and Prefect. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems solved.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Bengaluru, India

- **MLOps Pipeline – SME Program Graduate (Jun 2025 – Aug 2025):** Architected end-to-end SageMaker Pipelines platform for medical image classification – training, model registry, blue-green canary deployment with automatic rollback on error and latency alarms, auto-scaling, and PSI drift detection. Provisioned full infrastructure as code via Terraform + CodePipeline; exposed inference through API Gateway + Lambda with a CloudFront frontend. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0; see Publications).
- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (∼90%) building the AWS foundation for a data-centre exit and a second landing zone in Terraform: 4,000+ resources across 20+ Terraform Cloud workspaces, none failing.
  - Codified Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines.
  - Delivered two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect, plus self-service onboarding that gives each new workload its route table within 15 minutes.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a 10/10 CSAT.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC with Amazon Q.
- **Internal Impact:** Published a second AWS sample on Organizations governance (see Publications). Co-authored an AWS Prescriptive Guidance pattern on secure PR previews and approval-gated releases for Amplify Gen 2. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). Delivered 2 org-wide tech talks on Terraform testing and GitOps pipeline gating. Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for production observability.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS to embed continuous code quality gates into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Projects

Ledger Sync Jan 2026
React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

orrery Oct 2026
Agent Skill, Claude Code Plugin, Node.js, ELK, SVG GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Bedrock Multi-Model MCP Apr 2026
TypeScript, MCP Protocol, AWS Bedrock Converse API GitHub
◇ MCP server putting 26 text models from 7 providers, plus image, video and embedding models, behind 8 tools and one set of AWS credentials. Benchmarked in us-east-1: 26 of 26 models answered and 156 of 156 calls succeeded.

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

Other Notable Projects: itr-agent (deterministic tax engine, 151 tests), Claude Cost Optimizer (cache-aware Claude cost model), GroundZeroNN (neural network from scratch in NumPy), SelfHub (MCP personal-memory server), Instagram Autopilot (source-grounded animated SVG Reels on a GitHub Actions schedule; Bedrock writes the topic and caption).

Open Source Contributions (11 merged upstream)

- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

MLOps & ML Platform: SageMaker Pipelines, Model Registry, Endpoints (real-time & async), Custom PSI drift jobs, Blue-Green Deployment, A/B Traffic Shifting, Feature Stores (Feast)
ML Frameworks: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Experiment Tracking, Hyperparameter Tuning
CI/CD for ML: CodePipeline, CodeBuild, GitHub Actions, Docker, Model Versioning, Automated Retraining Triggers
Cloud & Infrastructure: AWS (Lambda, API Gateway, ECS Fargate, CloudFront, EventBridge, S3, CloudWatch), Terraform, HCL, tftest
Data & Serving: FastAPI, GraphQL, REST, Data Pipelines, SHA-256 Deduplication, SQLite, X-Ray, CloudWatch Metrics
Languages: Python, HCL, SQL, TypeScript, JavaScript, Bash